Jiezheng Wei

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(Updated: August 3, 2026)

Education

Ph.D., Economics, Singapore Management University	2022–2027 (Expected)
Thesis: AI and High-Frequency Analysis for Monetary Policy Shocks	
Committee: Jia Li (Chair)	
M.Sc., Economics, The Chinese University of Hong Kong, Shenzhen	2019–2021
B.Sc., Economics, Central University of Finance and Economics	2013–2017

Research Interests

Econometrics and AI Applications in Economics.

Working Papers

1. "News-jump Regressions: Weak Identification and Robust Inferences," with Jia Li.

   Abstract. News jumps, by which we mean asset-price jumps triggered by public news announcements, are increasingly attractive to empiricists because of their narrative interpretability and clean discontinuity-based identification. However, small news jumps raise weak-identification concerns when their signal-to-noise ratios are low, an issue highlighted as the "power problem" by Nakamura and Steinsson (2018). We introduce a notion of weak jumps, whose sizes are proportional to the square root of the length of the narrow time window around an event, to approximate empirically small announcement responses. We show that weak jumps render the "jump regressions" of Li, Todorov, and Tauchen (2017) inconsistently estimable and distort their regression-based inference. We develop identification-robust inferential methods based on the Anderson–Rubin and conditional approaches. In empirical applications, we invert the proposed tests to construct valid confidence intervals.

2. "A Fixed-k Bipower Jump Test," with Jia Li and Qiyuan Li.

   Abstract. We develop a jump test for Itô semimartingales observed at high frequency based on the ratio of untruncated realized variance to bipower variation within a block of k consecutive returns, where k is held fixed. Under infill asymptotics, the limiting distribution of the proposed test is asymptotically pivotal under the null of no jumps, according to a coupling result inherited from the driving Brownian diffusion, and therefore delivers favorable statistical properties over short time windows. Monte Carlo experiments calibrated to empirical financial data show that the proposed test attains essentially nominal size for small k. The framework provides a single, tuning-free, block-localized price jump detector that plugs directly into the broader apparatus of fixed-k volatility inference in Bollerslev, Li, and Liao (2021) and the bipower jump test of Barndorff-Nielsen and Shephard (2006).

3. "An Honest Inference for High-Dimensional Linear Models," with Dennis Lim, Jonathan HW Tan and Pengjin Min.

   Abstract. Inference in high-dimensional settings poses significant challenges. Building on the influential work of desparsified LASSO in van de Geer, Buhlmann, Ritov, and Dezeure (2014), we develop an inference procedure that extends their approach in two key directions: (i) we allow inference on linear combinations of the true parameter vector at moderate, potentially diverging rates; and (ii) we accommodate general heteroskedasticity in the data. Under mild regularity conditions, we demonstrate both theoretically and through simulations that our proposed test achieves asymptotically exact size control.

4. "Evaluating LLM forecasts on Sport Events," with Jia Li and Andrew Patton.

5. "Identifying Monetary Policy Shock with LLMs," with Sen Zhang.

Presentations

2025	SMU Students Econometrics Workshop.
2024	SMU Applied Micro Reading Group and SMU Students Econometrics Workshop.
2023	SMU Students Econometrics Workshop.
2021	International Conference on Smart Finance, CUHK-Shenzhen.

Research Experience

Research assistant for Prof. Jia Li, Singapore Management University, 2024 - 2026.

Teaching Experience

Singapore Management University

TA, Time Series Econometrics (PG), 2026

TA, Econometric Analysis (PG), 2025

TA, Intermediate Econometrics (UG), 2025

TA, Panel Data Analysis (UG), 2023
TA, Real Estate Economics (UG), 2023, 2025

The Chinese University of Hong Kong, Shenzhen
TA, Applied Probability and Stochastic Processes in Business (UG), 2022
TA, Machine Learning in Business (UG), 2021, 2022
TA, Basic Statistics (UG), 2021
TA, Advanced Econometrics (PG), 2020

Awards & Honors

Singapore Management University - China Scholarship Council Joint Fellowship, 2022 - 2026.
Presidential Award for Outstanding Master's Students, CUHK-Shenzhen, 2021.
Dean's List, School of Management and Economics, CUHK-Shenzhen, 2021.
First Class Scholarship for Academic Excellence and Dean's List, CUHK-Shenzhen, 2021.

Work Experience

2021-22	Research Associate, SME, The Chinese University of Hong Kong, Shenzhen.
2017-19	Associate, Beijing Dusheng Investment Fund (LP), Beijing, China.

Other Services

2024 - 25	Executive Committee, Graduate Research Student Society, SMU.

Others

AI skills: CLI-based AI (ChatGPT/codex, Claude), AI API.
Programming skills: Python, R, Matlab, $\LaTeX$.
Languages: English (fluent), Chinese (native).
Hobbies: fitness, badminton, classic music, cook.